

GATM Quiz 1.

Analysis Honors 25/26 Deggeller / Hlasek

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Period 5

No calculator. No notes or tables.

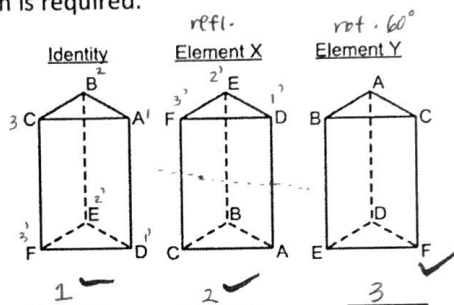
You can use a flip triangle, 2 snap strips, and 2 models (a triangular prism and a cube) with a stick.

1. An element of the 15-post snap group has period 105. What are the sizes of the components of this element? [2 pts]

$$\begin{array}{r} 1 \\ 35 \\ \times 3 \\ \hline 105 \end{array} \quad \begin{array}{r} 15 \\ 5 \quad 10 \\ \swarrow \quad \searrow \\ 7 \quad 3 \end{array}$$

Sizes of components: 5, 7, 3 ✓

2. Three elements of the rotation/reflection group of an equilateral triangular prism are drawn below. What is the period of each? No explanation is required.



Periods:

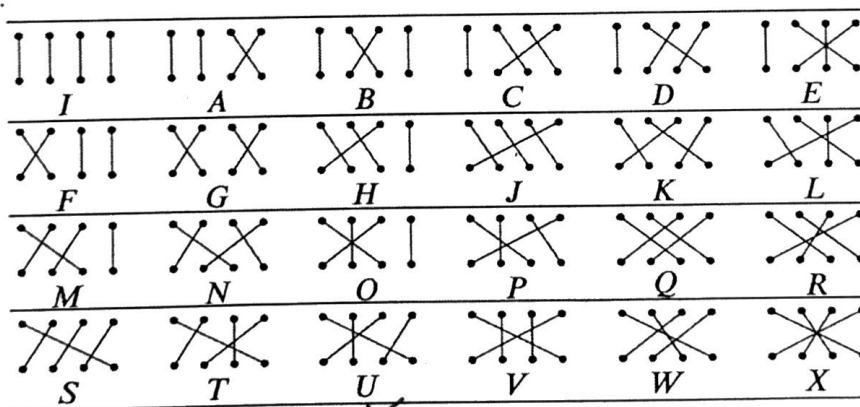
1

2 ✓

3 ✓

[1 pt each]

3. The 4-post snap group is shown below. Complete each blank using a single letter. No explanation is required.



- a) What is the inverse of T? L [1 pt] ✓

- b) Evaluate QM^2Q^2M = Q [1 pt] ✓

- c) Give an example of an element with period 3. (Many answers are possible, choose one.)

M [1pt]

- 2 x d) Give an example of a subgroup with 3 elements. (Many answers are possible, choose one.)

I, D, E [2 pts total] ✓

DoE=B not closed!

IXI

4. Decide about each statement below if it is always (A), sometimes (S), or never (N) true.

a) Any one element of a group can generate a group.

S ✓

b) The order of snap group S_n equals to $2n$. (for S_3)

S ✓

-1 c) A subgroup of a group has the same identity element as the group itself.

S ✗

5. Consider group $G = \{\text{all real numbers that are not equal } -\frac{1}{3}\}$ under binary operation $\boxtimes$ defined as $a \boxtimes b = a + b + 3ab$.

a) What is the identity in this group? [1 pt]

$$a \boxtimes I = I \boxtimes a = a$$

$$a + I + 3aI = a$$

$$I + 3aI = 0$$

$$I(1 + 3a) = 0$$

$$I = 0$$

b) Find the inverse of 7. Show your work. [2 pts]

$$7 \boxtimes y = 0$$

$$7 + y + 3(7)(y) = 0$$

$$7 + y + 21y = 0$$

$$22y = -7$$

$$y = -\frac{7}{22}$$

6. Group G is defined by the Cayley table to the right.

a) What is the identity in G ? 0 [1 pt]

b) What is the order of G ? 4 [1 pt]

c) What is the period of -1? 4 [1 pt]

d) What is the inverse of -2? -2 [1 pt]

	-2	-1	0	1
-2	0	1	-2	-1
-1	1	-2	-1	0
0	-2	-1	0	1
1	-1	0	1	-2

$$(-1) \vee (-1) = -2$$

$$(-1) \vee (-2) = 1$$

$$(-1) \vee (1) = 0$$

e) Is -2 a generator of G ? yes/no NO Justify your answer below. [2 pts]

with just -2, you can only generate -2 and 0; no matter how many times you perform the operation, you cannot generate all 4 elements

f) Group H is the set of rotations of a square. Are groups G and H isomorphic? If yes, provide one possible one-to-one correspondence by listing which element of H corresponds to each element of G . If not, justify why. [2 pts]

answer yes/no yes

	period	G
90°	4	0
180°	2	1
270°	4	-1
0°	1	-2

Complete only if G and H isomorphic:

-2 corresponds to rotation by 180°

-1 corresponds to rotation by 90°

0 corresponds to rotation by 0°

1 corresponds to rotation by 270°

$$1 \vee (-2) = -1$$

$$270 \quad 180^\circ \stackrel{?}{=} 90$$

$$90^\circ \quad 180^\circ$$

$$(-1) \vee (-2) = 1$$

$$90^\circ \quad 180^\circ \stackrel{?}{=} 270^\circ$$

7. Which of the following sets under the operations listed are groups? Circle all that apply. [2pts total]

- a) ☐ The set $\{1, -1, i\}$ under multiplication
 b) ☒ The set of even integers under addition
 c) ☒ The set of all non-zero rational numbers under multiplication
 d) ☒ The set of all real numbers under subtraction

$$\sqrt{2} - 1$$

$$(x-y) - z$$

closure
 associativity
 invertibility
 identity

$$(3-5) - 2$$

$$-2 - 2 = -4$$

$$3 - (5-2)$$

$$3 - 3 = 0$$

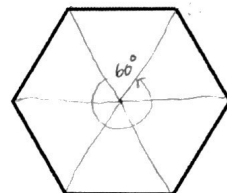
8. Which of the sets below are uncountable? Circle all that apply. [2pts total]

- a) ☒ powers of e with natural exponents $\{e^1, e^2, e^3, \dots\}$
 b) ☐ the set of all rational numbers
 c) ☒ the set of all complex numbers
 d) ☐ the set of all Choose Numbers $\binom{n}{k}$, where n and k are whole numbers

9. Consider the group H of rotations of a regular hexagon.

- a) The order of H is 6. [1 pt]
 b) Rotation by 120° has period 3. [1 pt]
 c) Does rotation by 300° generate the group? Explain your answer. [2pts]

yes - a rotation by 300° going one way (clockwise/ counterclockwise) is the same as a rotation the opposite way 60°. with 60°, you can generate every element by the time you rotate it 6 times.



Total quiz points : 29 /32

